

Research paper

Load-carrying performance and fatigue life of double-supported tapered roller bearing in offshore wind turbine

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ABSTRACT

The double-supported tapered roller bearing (DTRB) is a critical component of the transmission system in offshore semi-direct drive wind turbines, as its load-carrying performance directly affects operational efficiency and system reliability. In this study, an innovative method for evaluating the contact characteristics and fatigue life of the wind turbine main shaft DTRB is proposed. First, mechanical and finite element models of the main shaft DTRB system are established. Subsequently, the load-carrying mechanism and safety performance of the main shaft DTRB are investigated, followed by an analysis of the effects of different operating clearances on the contact characteristics. Finally, a full life-cycle load spectrum of the main shaft DTRB is constructed and the fatigue life is evaluated based on the Palmgren–Miner theory. The results demonstrate that the proposed method is both reliable and effective. Calculation results from the mechanical model of the main shaft DTRB show good agreement with experimental and finite element results. The tapered roller modification design significantly improves the safety performance. Additionally, the operating clearance range of the main shaft DTRB can be controlled between -270 and -150 μm , which can substantially improve its fatigue life.

1. Introduction

As global demand for renewable energy rises, wind power generation has attracted considerable attention (Wang et al., 2018). The Global Wind Energy Report 2024 states that the newly installed global wind power capacity reached a record 117 GW in 2023 (GWEC, 2024). Offshore wind power has become a critical component of global energy strategies owing to the superior offshore wind conditions (Dinh et al., 2022). However, the complexity of marine environments and inefficient maintenance strategies lead to high failure rates of key components in offshore wind power transmission systems, significantly affecting the economic returns of wind farms (Zheng and Chen, 2022; Liu et al., 2021; Guo et al., 2020). The double-supported tapered roller bearing (DTRB) is a critical component in the transmission systems of offshore semi-direct drive wind turbines, particularly in main shaft system applications (Fig. 1) (Gbashi et al., 2024; Touimi et al., 2018). Owing to complex and variable load conditions and extreme environmental factors, large tapered roller bearings are more prone to damage. Severe mechanical failures can lead to substantial downtime and maintenance costs, significantly compromising the operational reliability of wind turbines (Dhanola and Garg, 2020).

To enhance load-carrying performance and extend service life, extensive research on tapered roller bearings has been conducted. In terms of theoretical model development, Li and Gao (2022) studied the internal load distribution of DTRB in wind turbine main shafts by developing a five-degree-of-freedom mechanical model that considers preload, bearing clearance, raceway tilting, and roller profile. An iterative computational method was employed to determine the contact stress distribution between rollers and raceways. Building on this, Yuan et al. (2024) proposed a new method for calculating the internal load distribution that incorporates the axial gap based on the actual gap distribution assumption. However, the effect of preload in the main shaft DTRB system on bearing fatigue life was not analysed in detail. Jiang et al. (2022a,b) proposed a five-degree-of-freedom model that incorporates bearing clearance and preload and replaced the traditional Newton–Raphson method with a metaheuristic optimisation approach, significantly enhancing the convergence speed and efficiency. Zheng and Chen (2022) focused on double-row tapered roller bearing (TRB) widely used in offshore direct-drive wind turbine main shafts, introduced a comprehensive quasi-static model to analyse internal load and contact pressure distribution in double-row TRB, accounting for angular misalignment, combined external loads, and friction. The model

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revealed that even minor misalignment angles between the inner and outer rings can significantly alter the magnitude and distribution of contact forces and pressures. Zhang et al. (2019) focused on roller skewing and developed a model to calculate friction torque in dry-lubricated TRB, providing guidance for bearing design and torque optimisation. Tong and Hong (2017a, 2017b, 2017c) presented a design optimisation for the partially crowned roller profile to improve the TRB performance, offering a foundation for further geometric optimisation of rolling bearings. Furthermore, Cao et al. (2023) integrated lubrication analysis with mechanical studies, performing mixed lubrication analysis on TRB using numerical grinding methods and optimising the bearing profile. The results showed that the optimised profile enhanced performance by smoothing the contact pressure distribution, effectively reducing pressure peaks at roller ends and improving durability.

Time-varying heavy loads in wind turbine transmission systems significantly impact the fatigue life of main shaft bearings. Zheng et al. (2020a,b) proposed a novel integrated quasi-static model to study the fatigue life of double-row TRB under oscillating external loads and varying speeds, considering angular misalignment, internal clearance, and friction. Tong and Hong (2014, 2017) developed a stress distribution and fatigue life model for TRB by analysing displacement, contact load, stiffness, and load distribution, incorporating slice theory and a three-dimensional elastic contact model. They have also investigated the effect of misalignment angles on bearing life and operational torque (Tong and Hong, 2016). Kim et al. (2023) explored the thermo-mechanical behaviour of double-row TRB under thermal effects, revealing that interference fits rapidly degrade bearing life, underscoring the importance of optimising initial clearance values and roller end sphere radii. Li et al. (2021) proposed a wiener-based rolling bearing life prediction method using improved Kalman filtering and an adaptive modification algorithm, effectively improving stability and accuracy in rolling bearing life prediction. Golmohammadi et al. (2018) developed an efficient 3D elastic-plastic finite element model to characterise the rolling contact fatigue behaviour of through-hardened steel at high loads.

To overcome the limitations of traditional fatigue life prediction methods, Zhao et al. (2022) proposed a digital twin-based approach to predict main bearing life in wind turbines. This method integrated transfer learning and a convolutional long short-term memory network, effectively addressing data scarcity issues and improving prediction accuracy. Peeters et al. (2018) proposed a rolling bearing fault detection method based on vibration, effectively detecting the rolling bearing fault

of wind turbines and reducing the operation and maintenance cost of wind energy. Building on these theoretical models, Jiang et al. (2022, 2024) and Wu et al. (2023) introduced optimisation algorithms for bearing design, dynamic analysis, and lubrication performance evaluation.

Finite element analysis (FEA) is widely applied in offshore engineering machinery. Zhang et al. (2017) proposed a semi-analytical method for solving sliding finite-length space contact problems, considering the influence of the free end face, improving the efficiency of solving the contact stress. FEA enables accurate prediction and performance optimisation of wind turbines components via accurate modelling and analysis, thereby improving their reliability and safety (Jin et al., 2023; Wang et al., 2023; Liang, 2025). To enhance the accuracy of FEA results, Lostado et al. (2015, 2017) proposed a non-linear finite element model to address the high contact stress issue on the raceways of double-row TRB. The finite element model was refined and validated using theoretical models and pressure sensors.

The configuration of the main shaft DTRB has become the mainstream application trend in offshore semi-direct drive wind turbines. Current research mainly focuses on local contact analysis or theoretical calculation models of roller bearings. However, there is insufficient systematic modeling and verification of load distribution for the main shaft DTRB under actual wind load conditions, as well as on roller shape optimisation, bearing preload control, and the coupling mechanism of fatigue life. In particular, there is still a lack of research on the impact of the preload of the main shaft DTRB system on fatigue life under the full-life fatigue load spectrum of wind turbines. Therefore, this study aims to develop and validate a comprehensive performance evaluation method for the main shaft DTRB, facilitating quantitative predictions and design optimisation for contact behavior and fatigue life.

The innovation of this study lies in proposing an innovative evaluation method that can comprehensively consider the actual wind loads to conduct a systematic investigation into the load-carrying mechanism and fatigue life of the main shaft DTRB structure, thereby effectively guiding the design, optimising the contact behavior, and improving the fatigue life. The rest of this paper is organised as follows. In Section 2, a theoretical contact mechanics model for the wind turbine main shaft DTRB is developed and a fatigue life calculation model for the wind turbine main shaft DTRB is constructed based on the Palmgren-Miner theory. Section 3 establishes a finite element model of the wind turbine main shaft transmission system and a localised refined sub-model of the main shaft DTRB, integrating FEA technology, with its effectiveness

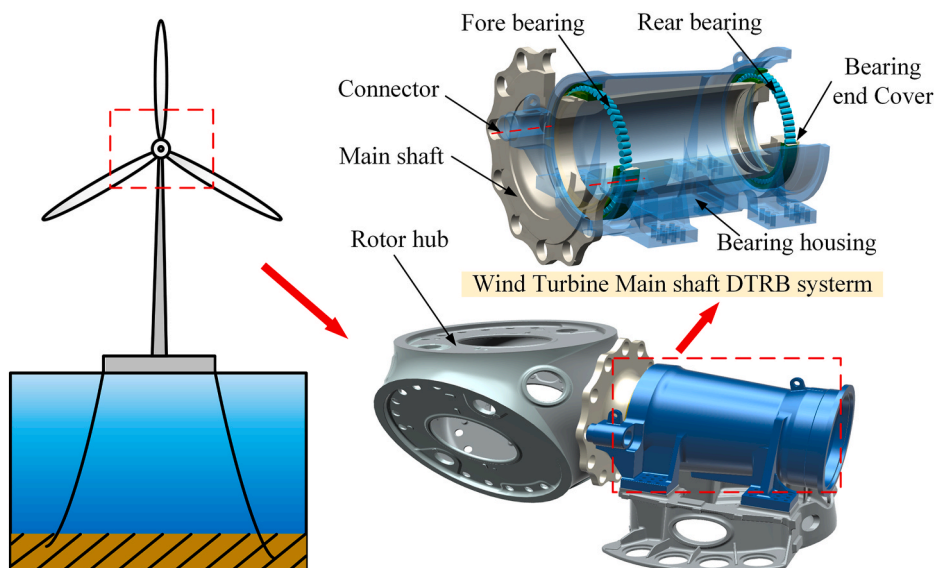


Fig. 1. Schematic of the main shaft DTRB system for the 6 MW semi-direct drive offshore wind turbine.

validated via experimental testing. Section 4 analyses the load-carrying mechanism of a 6 MW wind turbine main shaft DTRB under typical load conditions and evaluates the contact characteristic and safety performance of the main shaft DTRB. Finally, the fatigue life of the main shaft DTRB is calculated using the full life-cycle load spectrum of wind turbines, and a rational preload amount range for the main shaft DTRB is determined. Section 5 summarises the major research conclusions.

2. Mechanical model of the DTRB

This section presents a quasi-static mechanical model for the internal load distribution of the main shaft DTRB, considering the influence of bearing clearance. The structure of the TRB is shown in Fig. 2. It consists of four primary components: the inner ring, outer ring, roller, and cage. Here, D_w is the large-end diameter of the tapered roller, d_w is the small-end diameter, D_m is the mean diameter, L_{we} is the effective contact length, d_m is the pitch diameter, L_a is the axial clearance, and L_r is the radial clearance. $L_r = L_a \tan \alpha_e$, where α_e is the contact angle of the outer raceway, α_i is the contact angle of the inner raceway, α_m is the mean contact angle between the inner and outer raceways, ε is the semi-cone angle of the tapered roller, and α_f is the contact angle of the large rib on the inner ring.

For theoretical modelling, several assumptions are commonly made for simplification, considering the application characteristics of the wind turbine main shaft DTRB and the uncertainties arising from non-linear interactions between rollers and raceways (Yang et al., 2018; Zheng et al., 2020a,b):

- (1) The operating conditions involve low speed and heavy load. Therefore, the centrifugal force of the rollers is neglected in the model.
- (2) Based on Hertz contact theory, the roller and inner raceway are modelled as isolated bodies for mechanical analysis, with the outer raceway fixed in space. The materials of the bearing components are assumed to be homogeneous, and the linear elastic deformation between the roller and outer raceway is localised at the contact region. Thus, the structural deformation of the bearing is neglected.
- (3) The effects of friction, lubrication, temperature, and the cage on the rollers are ignored.

2.1. Deformation compatibility relationship

The large offshore semi-direct drive wind turbine main shaft DTRB configuration is typically installed back-to-back to enhance resistance to

overturning moments. The inner rings of the two bearing rows are treated as a single entity, whereas the two outer rings are considered separate. Thus, the overall displacement of the inner rings relative to the outer rings is defined by radial displacement δ_r , axial displacement δ_a , and angular displacement θ . The quasi-static mechanical model for the wind turbine main shaft DTRB is established based on the load-displacement relationship between the tapered rollers and raceways, as shown in Fig. 3.

After applying a contact load Q_{ej} to the tapered roller bearing, radial elastic deformation δ_r and axial elastic deformation δ_a occur in the contact region between the roller and raceway, as shown in Fig. 4(a). At position angle φ_j , the radial elastic deformation component of the roller is

$$\delta_{rj} = \delta_r \cos \varphi_j \quad (1)$$

The axial elastic deformation δ_a between the roller and raceway is identical for each roller. Therefore, the axial elastic deformation component of the bearing is given by

$$\delta_{aj} = \delta_a \quad (2)$$

Therefore, starting from the roller exhibiting the largest radial contact elastic deformation between the roller and the raceway (Fig. 4(b)), the total displacement at the j th roller position along the contact normal direction of the outer raceway is

$$\delta_{nj} = \delta_{rj} \cos \alpha_e + \delta_{aj} \sin \alpha_e \quad (3)$$

As shown in Fig. 4(c), assuming the distance between the centre of the roller and the bearing system centre O_i is R , with its projections in the radial and axial directions denoted as R_1 and R_2 , respectively. The relative rotation angle between the inner and outer rings, generated by the deflection angle at the j th roller, is

$$\theta_j = \theta \cos \varphi_j \quad (4)$$

Thus, the radial and axial displacement components of the bearing, resulting from the misalignment angle between the inner and outer rings, are derived as follows:

$$\begin{cases} \delta_{r0j} = R\theta_j \sin \beta = R_2\theta \cos \varphi_j \\ \delta_{a0j} = R\theta_j \cos \beta = R_1\theta \cos \varphi_j \end{cases} \quad (5)$$

where β is the angle between the radial direction of the bearing system centre and the line connecting the roller centre, R_1 is related to the pitch diameter of the bearing roller, and $R_1 = d_m/2$.

If an axial preload F_0 is applied during the installation of the paired bearings, the axial pre-deformation can be expressed as follows:

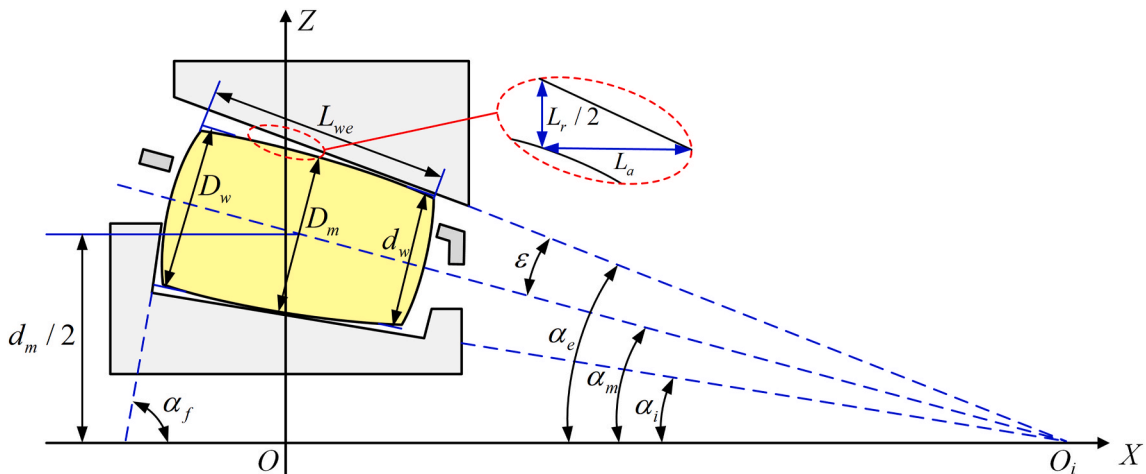


Fig. 2. Structure of tapered roller bearing.

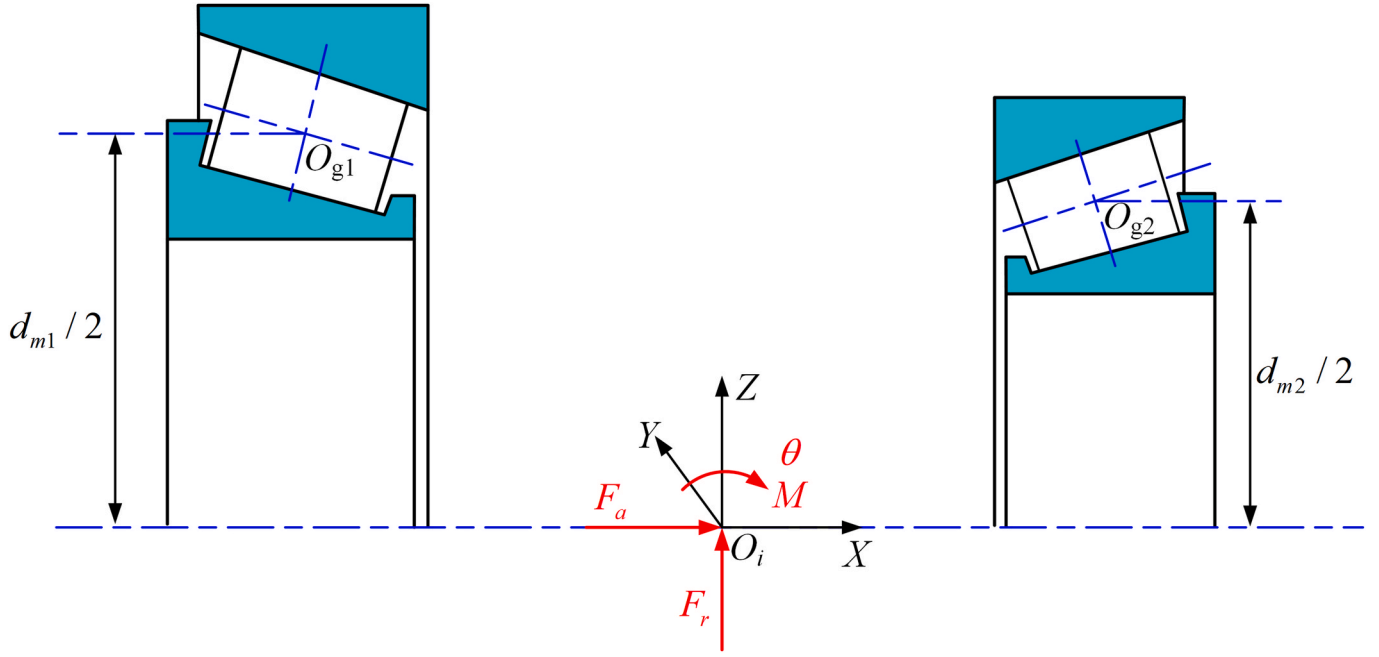
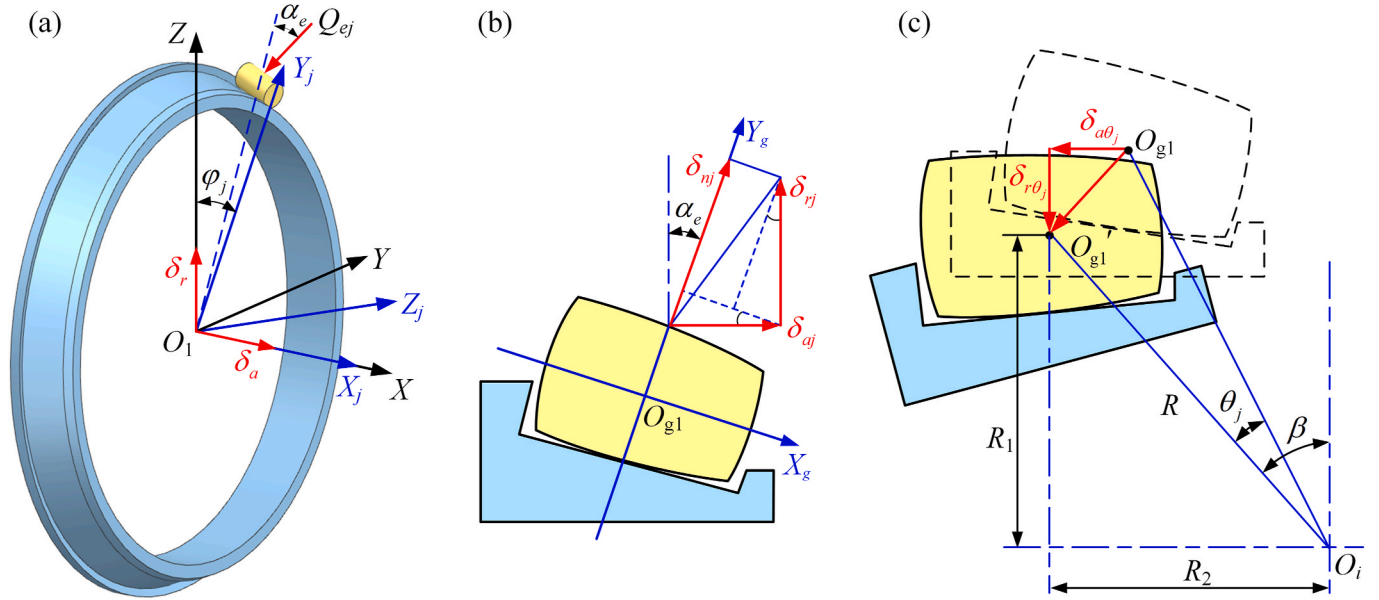


Fig. 3. Coordinate system of main shaft DTRB.

Fig. 4. Coordinate systems of the tapered roller bearing: (a) Roller position coordinate system, (b) Roller local displacement coordinate system, and (c) Displacement components of rotation angle θ .

$$\delta_0 = \{F_0 / [ZK(\sin \alpha_e)^{n+1}]\}^{1/n} \quad (6)$$

where F_0 is the axial preload, Z is the number of rollers, K is the load-deflection constant, and n is the load-deflection exponent. For a given roller-raceway contact, $n = 10/9$.

According to Eq. (1)–(6), the total radial and axial displacements at the j th roller, considering the bearing clearance, are expressed as follows:

$$\begin{cases} \delta_{rj} = \delta_r \cos \varphi_j + \delta_{r\theta j} - L_r/2 \\ \delta_{aj} = \delta_a + \delta_0 + \delta_{a\theta j} - L_a \end{cases} \quad (7)$$

2.2. Establishment of the equilibrium equation

The relationship between the elastic deformations of the rollers and raceways and the contact load is given by (Palmgren, 1959)

$$Q = K\delta_{nj}^n \quad (8)$$

$$K = 8.06 \times 10^4 L^{8/9} \quad (9)$$

Where Q is the contact load between the roller and raceway, K is the load-deflection constant, δ_{nj} is the contact deformation, n is the load-deflection exponent, for roller-raceway contact, $n = 10/9$, and L is the effective contact length of the roller and raceway.

The resisting moment generated by the contact load is

$$M_{cj} = M_{aj} + M_{rj} = 0.5(d_m \sin \alpha_e + d_c \cos \alpha_e) \sum_{j=1}^Z Q_{ej} \cos \varphi_j \quad (10)$$

where M_{aj} is the moment generated by the axial component of the contact load relative to the bearing centre O_i , M_{rj} is the moment generated by the radial component of the contact load relative to the bearing centre O_i , d_c is the distance between the centres of the two bearing rows, and $d_c = 2R_2$.

According to Eq. (3), (7–10), the mechanical equilibrium equation is established as follows:

$$\mathbf{F}_q = \begin{bmatrix} F_r - \sum_{i=1}^2 \sum_{j=1}^Z Q_{eij} \cos \alpha_{ei} \cos \varphi_{ij} \\ F_a - \sum_{i=1}^2 \sum_{j=1}^Z Q_{eij} \sin \alpha_{ei} \\ M - \sum_{i=1}^2 \sum_{j=1}^Z M_{cij} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

where i is the row number of the corresponding bearing, and j is the number of the corresponding roller.

The loading condition of the roller is shown in Fig. 5. The contact loads between the j th tapered roller and the inner raceway, outer raceway, and rib are Q_{cj} , Q_{ej} , and Q_{fj} , respectively, with contact angles α_i , α_e , and α_f , respectively. After achieving force equilibrium for the roller, the following relationship is satisfied:

$$\begin{cases} Q_{ej} \sin \alpha_e - Q_{cj} \sin \alpha_i - Q_{fj} \sin \alpha_f = 0 \\ Q_{ej} \cos \alpha_e - Q_{cj} \cos \alpha_i + Q_{fj} \cos \alpha_f = 0 \end{cases} \quad (12)$$

Considering the actual inclination angle of the main drive system of the wind turbine, the force analysis of the main shaft DTRB system under wind load at the hub rotor centre is illustrated in Fig. 6.

As shown in Fig. 6, O denotes the hub rotor centre, γ is the installation inclination angle of the main shaft, G_1 is the gravitational force of the main shaft, and G_2 is the gravitational force of the gearbox. L_1 is the distance from the fore bearing to the hub centre, L_2 is the distance from the centre of gravity of the main shaft to the hub centre, L_3 is the distance from the rear bearing to the hub centre, and L_4 is the distance from the centre of gravity of the gearbox to the hub centre. The mechanical equilibrium equation of the main shaft transmission system can be determined as

$$\begin{bmatrix} F_x \\ F_y \\ F_z \\ M_y \\ M_z \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -L_1 & -L_3 \\ 0 & L_1 & L_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} F_{xa} \\ F_{y1} \\ F_{y2} \\ F_{z1} \\ F_{z2} \end{bmatrix} = \begin{bmatrix} -(G_1 + G_2) \sin \gamma \\ 0 \\ (G_1 + G_2) \cos \gamma \\ -(G_1 L_2 + G_2 L_4) \cos \gamma \\ 0 \end{bmatrix} \quad (13)$$

where F_x , F_y , F_z , M_y , and M_z are the time-series load data at the centre of

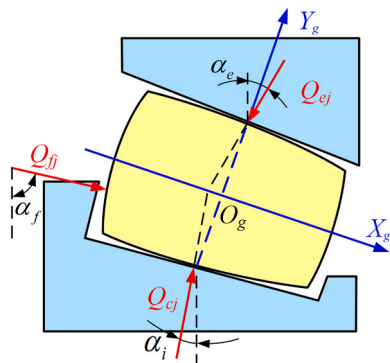


Fig. 5. Loading condition of the tapered roller.

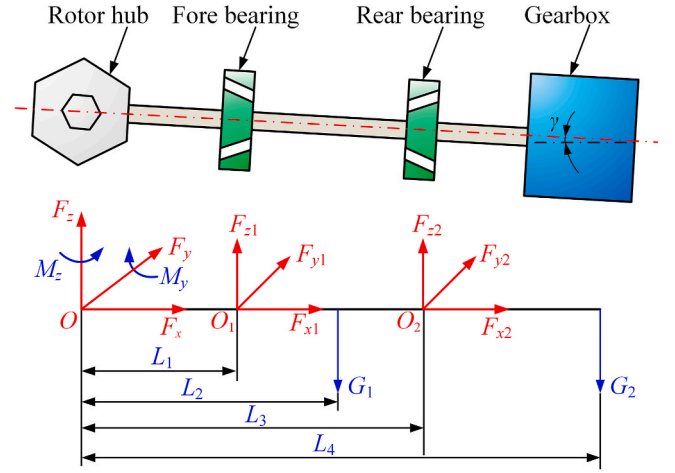


Fig. 6. The force analysis of the main shaft DTRB system.

the hub. F_{xa} is the resultant axial force generated by the external wind load. F_y and F_z are the radial forces acting on the fore and rear bearings along the y -direction and z -direction, respectively.

According to Eq. (13), the forces acting on the fore and rear bearings are given as follows:

$$\begin{cases} F_{ri} = \sqrt{F_{yi}^2 + F_{zi}^2}, (i = 1, 2) \\ F_{xa} = F_{x1} + F_{x2} \end{cases} \quad (14)$$

where F_r and F_{xa} are the resultant radial and axial forces generated by the external wind load. F_{x1} and F_{x2} are the axial forces acting on the fore and rear bearings along the x -direction, respectively.

The fore and rear bearings collectively support the axial load induced by the external wind load in the main shaft system, thereby maintaining the stability and balance of the bearing system in the axial direction. The derived axial force generated by the main shaft DTRB under radial load is given by

$$F_{di} = \frac{F_{ri}}{2Y_i}, (i = 1, 2) \quad (15)$$

where F_{d1} and F_{d2} are the derived axial forces generated by the fore and rear bearings, respectively. The axial load coefficient Y is provided by the bearing manufacturer.

Based on a comprehensive assessment of the wind load at the hub rotor centre, the load-carrying state of the main shaft DTRB is determined (Yuan et al., 2024). If $F_{d2} + F_{xa} > F_{d1}$, the fore bearing is compressed and the following relationship holds:

$$\begin{cases} F_{x1} = F_{d2} + F_{xa} \\ F_{x2} = F_{d2} \end{cases} \quad (16)$$

If $F_{d2} + F_{xa} \leq F_{d1}$, the rear bearing is compressed and the following relationship holds:

$$\begin{cases} F_{x1} = F_{d1} \\ F_{x2} = F_{d1} - F_{xa} \end{cases} \quad (17)$$

2.3. Contact pressure equations

The contact pressure between the roller and the raceway significantly influences the load-carrying state of the bearing. In the (x, y) coordinate system, the normal contact elastic deformation between two elastic bodies under frictionless conditions is determined as (Boussinesq, 1885)

$$w(x, y) = \frac{1}{\pi E} \iint_S \frac{p(x', y') dx' dy'}{\sqrt{(x - x')^2 + (y - y')^2}} \quad (18)$$

where $p(x', y')$ is the contact pressure, S is the contact region between the roller and raceway, and the parameter E' is given by

$$\frac{1}{E'} = \frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \quad (19)$$

where ν_1 and ν_2 and E_1 and E_2 are the Poisson's ratios and elastic moduli of the two contacting bodies, respectively.

The total normal contact elastic deformation can be obtained by integrating the contact pressure distribution $p(x', y')$ within the contact region S , as given by the following equation:

$$\iint_S p(x', y') dx dy = Q, p(x', y') \geq 0 \quad (20)$$

$$w(x, y) = \delta - z(x, y) - h \quad (21)$$

where Q and $w(x, y)$ are the normal contact load and elastic deformation between the roller and raceway, respectively.

According to Eqs. (18) and (21), the following relationship can be obtained:

$$\frac{1}{\pi E} \iint_S \frac{p(x', y') dx' dy'}{\sqrt{(x - x')^2 + (y - y')^2}} = \delta - z(x, y) - h \quad (22)$$

where δ is the initial displacement value between the two contacting bodies, $z(x, y)$ is the initial spacing between the roller and raceway at coordinate (x, y) , and h is the modification amount resulting from the roller profile modification.

The edge stress concentration resulting from the loading of tapered rollers causes the actual contact stress between the roller and raceway to significantly exceed the allowable Hertz contact stress (Li and Gao, 2022). The roller profile is commonly modified to alleviate this effect, as shown in Fig. 7(a). Based on slice theory, the contact region between the roller and the raceway is divided into n slice elements. Considering the logarithmic curve modification profile of the roller, the modification amount $h_{(k)}$ for the k th slice element is given by (ISO/TS 16281, 2008)

$$h_{(k)} = 0.00045 D_m \ln \left(\frac{1}{1 - (2l_k / L_{we})^2} \right), -L_{we} / 2 \leq l_k \leq L_{we} / 2 \quad (23)$$

where D_m is the mean diameter of the roller, l_k is the position of the k th

slice element, and Δl_n is the width of each slice.

The radius of curvature near the contact point for the cross-section of each slice element of the tapered roller is given by (Zheng et al., 2020a, b)

$$R_j = \frac{D_j}{2 \cos \varepsilon} \left(\frac{1 + 2 \sin^2 \varepsilon + 4 \sin^4 \varepsilon / \cos 2 \varepsilon}{1 + \tan \varepsilon \tan 2 \varepsilon} \right) \quad (24)$$

where D_j is the cross-sectional diameter perpendicular to the roller centreline and ε is the semi-cone angle of the roller.

As shown in Fig. 7(b), the geometric relationship between the contact half-width, the radius of curvature, and the initial displacement value for the tapered roller and the raceway is (Palmgren, 1959)

$$\begin{cases} \delta = R_j - \sqrt{R_j^2 - a_j^2} = R_j \left(1 - \sqrt{1 - (a_j / R_j)^2} \right) \\ 1 - \sqrt{1 - (a_j / R_j)^2} \approx 1 - \frac{a_j^2}{2 R_j^2} \\ \delta = 3.81 \left(\frac{1 - \nu_1^2}{\pi E_1} + \frac{1 - \nu_2^2}{\pi E_2} \right)^{0.9} \frac{Q^{0.9}}{L_{we}^{0.8}} \end{cases} \quad (25)$$

where R_j is the radius of curvature, a_j is the contact half-width of the tapered roller, and δ is the initial displacement value. ν_1 and ν_2 and E_1 , and E_2 are the Poisson's ratios and elastic moduli of the roller and raceway, respectively.

As shown in Fig. 7(c), to simplify the contact problem analysis, the potential contact region is divided into n strip elements along the roller axis. Assuming that the contact stress within element j is uniformly distributed along the y -axis and follows the Hertz distribution along the x -axis, the relationship is

$$\begin{cases} p_j = p_{oj} \sqrt{1 - (x/a_j)^2} \\ a_{oj} = \frac{2 R_j}{E} p_{oj} \\ w_{ij} = \frac{p_{oj}}{\pi E} \int_{-a_j}^{a_j} \int_{y_j - \Delta l_{nj}/2}^{y_j + \Delta l_{nj}/2} \frac{\sqrt{1 - (x'/a_j)^2} dx' dy'}{\sqrt{(x - x')^2 + (y_i - y_j - y')^2}} \end{cases} \quad (26)$$

where p_{oj} is the maximum contact stress on slice element j and w_{ij} is the displacement generated by contact stress p_{oj} at the centre of element i .

According to Eqs. (23)–(26), Eqs. (20) and (22) are given as follows:

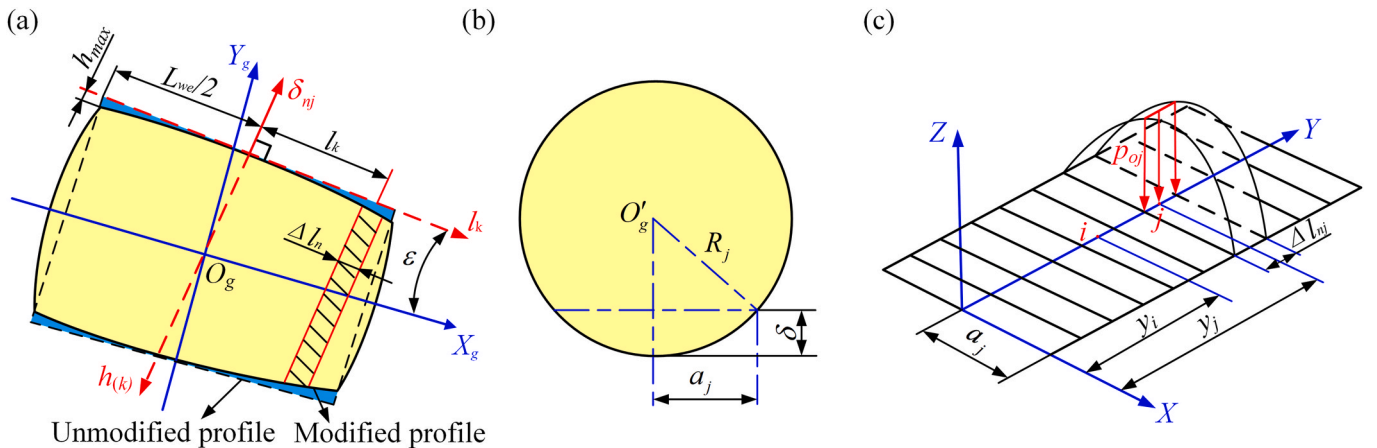


Fig. 7. Schematic of the contact between the roller and raceway: (a) Schematic of roller normal deformation and profile, (b) Geometric relationship between the contact half-width and initial displacement value, and (c) Discretisation of the contact region.

$$\begin{cases} \pi \sum_{j=1}^n a_j \Delta l_{ij} p_{oj} = Q, p_j \geq 0, (j = 1, 2, \dots, n) \\ \frac{1}{\pi E'} \sum_{j=1}^n D_{ij} p_{oj} = \delta - z(x, y) - h_j, (j = 1, 2, \dots, n) \end{cases} \quad (27)$$

where the calculation of the flexibility coefficient D_{ij} is given by

$$D_{ij} = \int_{-a_j}^{a_j} \int_{y_j - \Delta l_{ij}/2}^{y_j + \Delta l_{ij}/2} \frac{\sqrt{1 - (x'/a_j)^2}}{\sqrt{(x')^2 + (y_i - y_j - y')^2}} dx' dy' \quad (28)$$

2.4. Fatigue life assessment

A comprehensive assessment of the fatigue life of the main shaft DTRB in wind turbines is crucial for ensuring safe and reliable operation. The basic dynamic load ratings for the inner and outer raceways of the main shaft DTRB are given as

$$\begin{cases} Q_{ci} = b_m 552 \lambda \frac{(1 - \eta)^{29/27}}{(1 + \eta)^{1/4}} \left(\frac{\eta}{\cos(\alpha_m)} \right)^{2/9} \left(\frac{D_w + d_w}{2} \right)^{29/27} L_{we}^{7/9} z^{-1/4} \\ Q_{ce} = b_m 552 \lambda \frac{(1 + \eta)^{29/27}}{(1 - \eta)^{1/4}} \left(\frac{\eta}{\cos(\alpha_m)} \right)^{2/9} \left(\frac{D_w + d_w}{2} \right)^{29/27} L_{we}^{7/9} z^{-1/4} \end{cases} \quad (29)$$

where $\eta = [(D_w + d_w) \cdot \cos \alpha_m] / 2d_m$ is the basic dynamic load rating coefficient with a value of 1.1 and λ is the reduction factor related to stress concentration, with a value ranging from 0.6 to 0.8.

The inner raceway of the main shaft DTRB is a rotating component. The equivalent dynamic loads for the rotating raceway and non-rotating raceway are given as follows (ISO/TS 16281, 2008):

$$\begin{cases} Q_{ei} = \left(\frac{1}{z} \sum_{j=1}^z Q_{ij}^{4.5} \right)^{1/4.5} \\ Q_{ee} = \left(\frac{1}{z} \sum_{j=1}^z Q_{ej}^4 \right)^{1/4} \end{cases} \quad (30)$$

The average equivalent load of the main shaft DTRB over its full life-cycle is given by (Germanischer, 2010)

$$F = \left(\frac{\sum F_i t_i}{T} \right)^{\frac{1}{p}} \quad (31)$$

where t_i is the operating time under the equivalent dynamic load F_i , T is the design life of the main shaft DTRB, and $p = 10/3$ for roller bearings.

The cumulative fatigue failure of the main shaft DTRB under each load level can be evaluated based on the Palmgren–Miner theory, expressed as

$$\sum_{i=1}^n \frac{n_i}{N_i} = \sum_{i=1}^n \frac{q_i \cdot L_{10mr}}{L_{10mri}} = 1 \quad (32)$$

where n_i is the number of loading cycles at the i th load level, N_i is the number of loading cycles causing fatigue failure at the i th load level, q_i is the time proportion of the i th load level, and L_{10mri} is the modified rating life of the bearing on the i th load level for the main shaft DTRB.

The combined modified rating life L_{10mr} of the main shaft DTRB is given by

$$L_{10mr} = \frac{1}{\sum_{i=1}^n \frac{q_i}{L_{10mri}}} \quad (33)$$

where, $L_{10mri} = \left((Q_{ci}/Q_{ei})^{-9/2} + (Q_{ce}/Q_{ee})^{-9/2} \right)^{-8/9} 10^6 \text{ rev}$.

The structure of this paper is shown in Fig. 8, whose details are as follows:

Step 1: Input wind turbine hub rotor loads $\{F_x, F_y, F_z, M_x, M_y, M_z\}$, the structural parameters of the main shaft transmission system $\{L_1, L_2, L_3, L_4, G_1, G_2, \gamma\}$, and the structural parameters of the main shaft DTRB $\{d_m, D_w, d_w, L_{we}, L_{as}, z, \alpha_i, \alpha_e\}$.

Step 2: Use Eq. (13) to determine the load relationship of the main shaft DTRB, Eq. (15) to determine the derived axial force generated by the spindle DTRB under radial force, and Eqs. (16) and (17) to determine the bearing state of the main shaft DTRB.

Step 3: Assume initial guess values and convergence accuracy for the displacement of the main shaft DTRB $\{\delta_{r1}, \delta_{r2}, \delta_{a1}, \delta_{a2}, \epsilon_e\}$, compute the contact load value of the main shaft DTRB using Eq. (8), determine whether the load value convergence accuracy $\epsilon_e = 1 \times 10^{-2}$ is met, and adjust the deformation component $\{\delta_{r1}, \delta_{r2}, \delta_{a1}, \delta_{a2}\}$ until the force balance of Eq. (11) is applied. The load distribution of the main shaft DTRB is output.

Step 4: Based on the load distribution of the main shaft DTRB, input the load value Q of the maximum bearing roller. Set the convergence accuracy $\epsilon_1 = 1 \times 10^{-2}$ for the contact half-width and the convergence accuracy $\epsilon_2 = 1 \times 10^{-2}$ of the load value.

Step 5: Compute the total normal contact deformation $w(x, y)$ at each slice element using Eq. (21), and calculate the modification amount h_k at each slice element using Eq. (23). Compute the curvature radius R_j , initial displacement value δ using Eq. (24) and slice half-width $\Delta l_n = L_{we}/2n$. Compute the contact half-width a_j of each slice using Eq. (25).

Step 6: Compute the stiffness matrix coefficient D_{ij} using Eq. (28). Calculate the maximum contact stress p_{oj} at the centre of each slice using Eq. (26), and calculate the new contact half-width a_{oj} using Eq. (26). Check whether both the contact stress and contact half-width satisfy the following convergence conditions simultaneously: $p_{oj} \geq 0, |a_j - a_{oj}| \leq \epsilon_1$. If the above convergence conditions are not satisfied, set $p_{oj} = 0, a_{oj} = 0, a_j = 0.5(a_{oj} + a_j)$ and repeat step 6. If the above conditions are satisfied, proceed to step 7.

Step 7: Compute the load value Q_1 using Eq. (27) and then check whether the contact load value Q_1 satisfies the convergence condition $|Q_1 - Q|/Q \leq \epsilon_2$. If the above convergence condition is not satisfied, adjust δ and re-execute step 5. If satisfied, end the calculation and output each slice element's contact half-width and contact stress.

Step 8: Verify the accuracy of the contact analysis by comparing the theoretical contact stress values with the experimental and finite element values, as shown in Fig. 12.

Step 9: Using the mechanical model and the refined finite element sub-model, study the carrying mechanism of the main shaft DTRB under typical load cases and the effect of different operating clearances on the contact characteristics of the main shaft DTRB and evaluate the safety performance of the main shaft DTRB under extreme conditions.

Step 10: Based on the load distribution of the main shaft DTRB obtained in Step 3, compute the fatigue life of the main shaft DTRB under the full lifetime load spectrum of the wind turbine using Eqs. 27–31.

3. Finite element method of the DTRB system

3.1. Finite element modelling

According to the Germanischer Lloyd (GL) guidelines (Germanischer, 2010), the effect of the wind blade on the hub is fully simulated by determining the hub centre point O , as shown in Fig. 9(a). Boundary coupling constraints are applied to the three connection planes A between the hub and the blades. The wind load is applied at the geometric centre point O of the hub in the fixed coordinate system. A fixed constraint is applied to the connection plane B between the main frame and the wind turbine tower to simulate the bolted connection. A bonded constraint is also applied to connection plane C between the hub and the main shaft, the main frame and the bearing housing, the bearing and the end cover. No relative motion is assumed between the contact surfaces, and the effect of connecting bolts on force transmission is neglected.

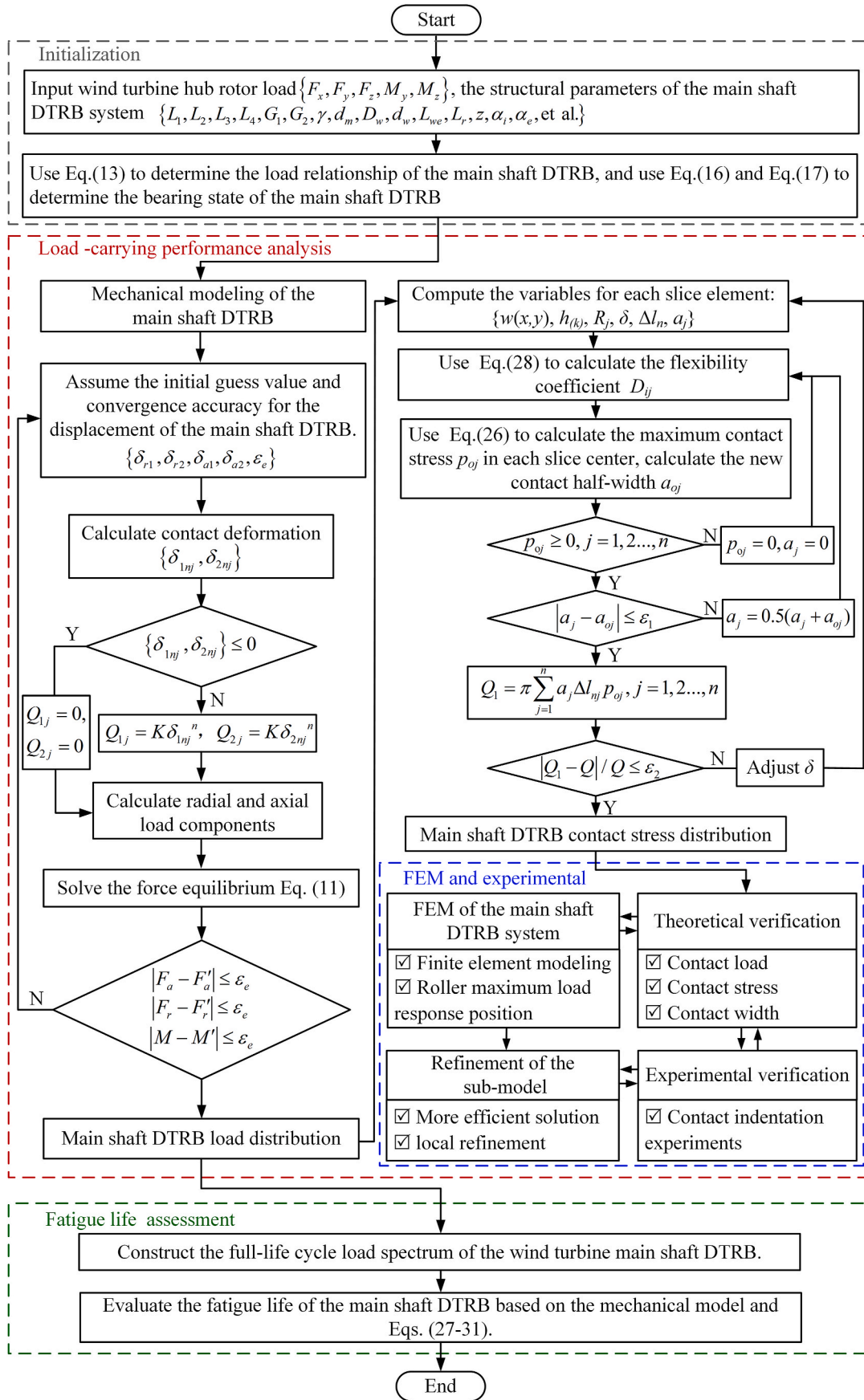


Fig. 8. Research flowchart of the study.

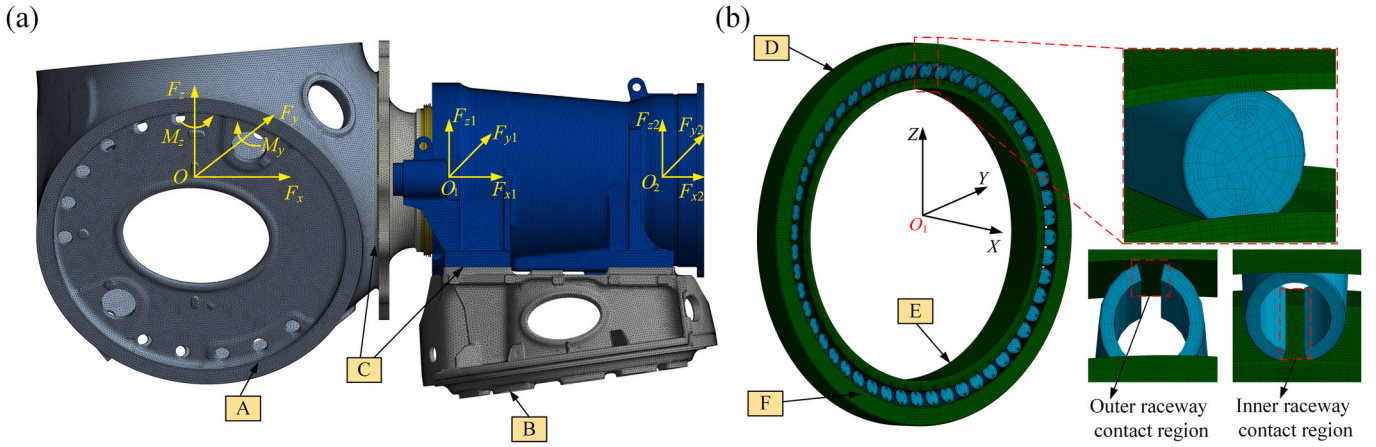


Fig. 9. (a) Main shaft DTRB system mesh model and boundary conditions and (b) Bearing roller and raceway mesh model.

The rollers of the fore and rear bearings are modelled using solid elements. The contact analysis between the rollers and raceways is a typical non-linear problem, where appropriate contact definitions are essential. The mesh size within the contact region and the boundary conditions are critical parameters for accurately predicting contact stress in the finite element model. A reduced contact region and coarse mesh size result in fewer nodes in the contact region, leading to inaccurate contact stress calculations. Accurate results are achieved when the number of contact elements along the contact half-width direction between the roller and raceway is approximately 10 (Lostado et al., 2015). To enhance the accuracy and convergence of the finite element model analysis for the main shaft DTRB system, local refinement is applied to the contact regions between the rollers and raceways, as shown in Fig. 9(b). The outer ring D of the main shaft DTRB is bonded to the bearing housing, and the inner ring E is constrained to rotate solely about the x-axis. The bearing end face F and the positioning mating surface in the shaft system are modelled with frictional contact, with a

friction coefficient of 0.3. The roller surface is the contact surface and the raceway surface is the target surface. Frictional contact is defined between the rollers and the inner/outer raceways with a friction coefficient of 0.15. A Fixed Joint in ANSYS Mechanical simulates the constraint of the cage on the roller.

3.2. Refinement of the sub-model

Improving the accuracy of the finite element model of the main shaft DTRB system will significantly increase the computational cost. To analyse the contact characteristics between the DTRB rollers and raceways more efficiently, a refined sub-model of the main shaft DTRB rollers and raceways is established in this section, as shown in Fig. 10.

As shown in Fig. 10(a), the solution is first computed in the finite element system model analysis module for the main shaft DTRB. Based on the results, the positions of the rollers and raceways corresponding to the maximum load response of the main shaft DTRB under actual wind

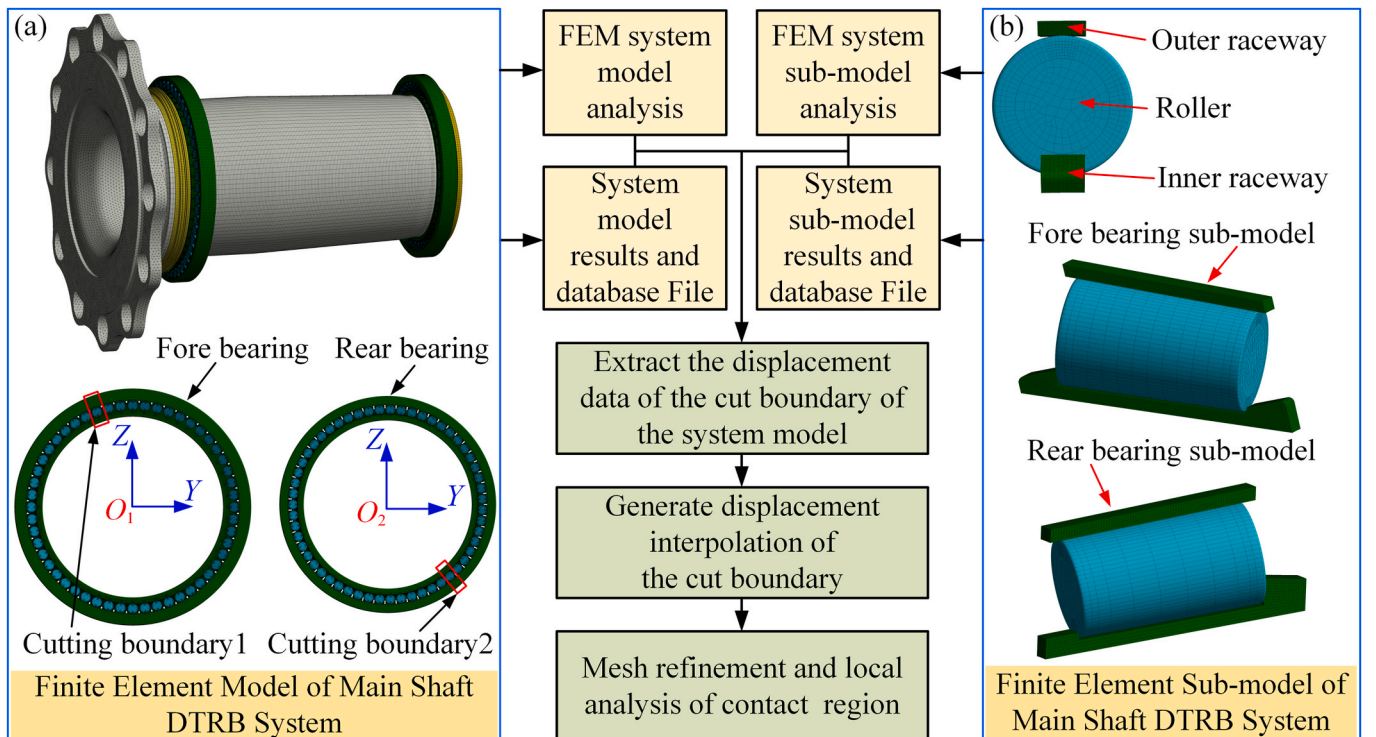


Fig. 10. Finite element mesh refinement of the main shaft DTRB: (a) System model and (b) Sub-model.

loads are determined. The main shaft DTRB is then segmented along Cutting Boundary 1 and Cutting Boundary 2. The results and database files of the finite element system model are saved, and the node displacement data for the two cutting boundaries are extracted. As shown in Fig. 10(b), the solution results of the main shaft DTRB finite element system model are read as the initial condition for sub-model analysis. The node displacement data from the cutting boundaries of the main shaft DTRB finite element system model are then interpolated and applied to the sub-model boundaries. A refined sub-model for the rollers and raceways is established, with the mesh division controlled by global mesh size and local refinement methods. To ensure that mesh quality does not affect the results, the mesh in the contact region between the rolling elements and raceways is further refined. The number of meshes between the roller and raceway in the main shaft DTRB sub-model is 35,689, with hexahedral mesh division adopted. The mesh size of the key contact region between the roller and raceway is set to 0.1 mm.

4. Results and discussion

This study investigates the load-carrying performance and fatigue life of the main shaft DTRB in a 6 MW wind turbine. The main shaft is supported by two single-row TRBs, with structural parameters provided in Table 1. A back-to-back arrangement increases the bearing span, reducing the load on the bearings and enhancing the resistance of the system to overturning moments. A mechanical model of the main shaft DTRB for offshore semi-direct drive wind turbines is established to analyse the load-carrying state under actual load cases. The contact characteristics and safety performance of the main shaft DTRB are analysed based on the mechanical model and refined finite element sub-model. The validity of the mechanical model and FEM is verified via indentation experiments. Finally, the fatigue life of the main shaft DTRB is evaluated based on the fatigue load spectrum.

Goodelle et al. (1970) applied the indentation method to study the contact problem in cylindrical rollers. The indentation method is an effective technique for characterising contact stress distribution and has been applied to measure occlusal forces in medical dentistry and contact indentation in bearings (Wang et al., 2022; Chen et al., 2023; Wang et al., 2022). The principle involves using a single-layer polyester film coated with a colour-developing substance and microcapsules containing colour-forming agents. When pressure is applied, the microcapsules rupture, releasing the colour-forming agents, which react with the colour-developing substance to produce red zones on the film. The size and shape of the contact region can be determined by measuring the indentation size.

To validate the accuracy of the proposed theoretical and finite element models, 42CrMo bearing steel was selected to prepare test specimens for the fore bearing sub-model. Contact indentation experiments between the tapered roller and raceway of the fore bearing under different load levels were conducted using a WDW-200E microcomputer-controlled electronic universal testing machine. As shown in Fig. 11(a) and (b), the experimental setup primarily consists of a loading

control system, hydraulic loader, outer ring, inner ring, tapered roller, and pressure measurement film. The loading control system can apply a load ranging from 1 to 200 kN and a speed from 0.001 to 500 mm/min.

Quasi-static loading tests were conducted to establish the relationship between load value and contact half-width and to validate the theoretical and finite element models. Tests were performed at load levels of 30 kN, 50 kN, 70 kN, 90 kN, 110 kN, 130 kN, 150 kN, and 170 kN, at 24 °C and 50 % humidity. The loading and unloading rate was 0.5 mm/min. Part of the indentation test results are shown in Fig. 11(c). To ensure the accuracy of the tests, each sample at every load level underwent three loading tests, and the average of the three indentation width measurements was used as the experimental data.

The experimental contact width was compared with the theoretical model. As shown in Fig. 12, the theoretical and experimental contact widths between the roller and raceway increase with the load. However, the error between the theoretical and experimental values is more pronounced at low load levels. This is because, during the initial contact stage between the roller and raceway, elastic deformation and complete contact leading to indentation occur only under sufficient external load. The roller and raceway may not have fully entered the contact stage at lower load levels, leading to larger errors. As the load increases, the error fluctuates but remains within 10 %. The results also show that the finite element values for the contact width between the tapered roller and raceway closely match the experimental values, with a maximum error of 5.8 %. This confirms the accuracy of the finite element sub-model for the main shaft bearing system.

4.1. Analysis of the carrying mechanism

During wind turbine operation, the main shaft DTRB experiences a wide range of load conditions (Hart et al., 2019). The structural data of the main shaft transmission system are provided in Table 2. Four typical load cases with high time proportions at the hub centre in the GL coordinate system are selected (Table 3). Based on the mechanical model, the load-carrying mechanism of the main shaft DTRB under typical load cases is investigated, as shown in Fig. 13.

As shown in Fig. 13(a)–(c), under case A, the overturning moment M_z causes the main shaft to tilt, altering the load-carrying region of the main shaft DTRB. Taking the first roller as the 0° reference, the load-carrying angle of the fore bearing is 246° clockwise, covering rollers 1 to 13 and 41 to 52, totalling 25 rollers. The load-carrying angle of the rear bearing is 55° clockwise, covering rollers 1 to 55, totalling 55 rollers. In this case, the fore bearing is relaxed and the rear bearing is compressed and fully loaded. Under case B, as the bending moment load M_y increases and the number of load-carrying rollers in both the fore and rear bearings remains constant, with the rear bearing staying compressed. As shown in Fig. 13(b)–(d), compared to case A, the maximum contact load of the fore bearing increases by 22.7 % and that of the rear bearing increases by 47.9 %. The introduction of the bending moment load M_y increases the load values of the main shaft DTRB, with a significant impact on the load-carrying performance of the rear bearing. Under case C, as the radial load F_z increases, the number of load-carrying rollers in both the fore and rear bearings remains constant, and the main shaft DTRB load values continue to increase proportionally. The maximum contact load of the fore bearing increases by 82.1 % and that of the rear bearing increases by 74.5 %. The rear bearing remains compressed at this point; however, the load on the fore bearing increases significantly, taking on a dominant role in supporting the radial load. Under case D, the fore bearing has 52 load-carrying rollers, reaching a fully loaded state with a maximum load decrease of 2.0 %. The rear bearing has 27 load-carrying rollers, experiencing a maximum load decrease of 16.6 %. In this case, the fore bearing is compressed and the rear bearing is relaxed. It can be seen from Fig. 13(b)–(d) that the maximum contact load values between the rollers and raceways of the main shaft DTRB under case D are significantly reduced. This is because of the contact angle of the TRBs, where part of the axial force generates a

Table 1
Structural parameters of the main shaft DTRB.

Parameter	Numerical value	
	Fore bearing	Rear bearing
Pitch diameter d_m (mm)	1731	1561
Diameter of roller large-end D_w (mm)	96	81
Diameter of roller small-end d_w (mm)	90	79
Roller effective contact length L_{we} (mm)	157	137
Number of rollers Z	52	55
Inner race contact angle α_i (°)	19	12
Outer race contact angle α_o (°)	21	13
Elastic modulus E (Gpa)	210	210
Poisson's ratio ν	0.3	0.3

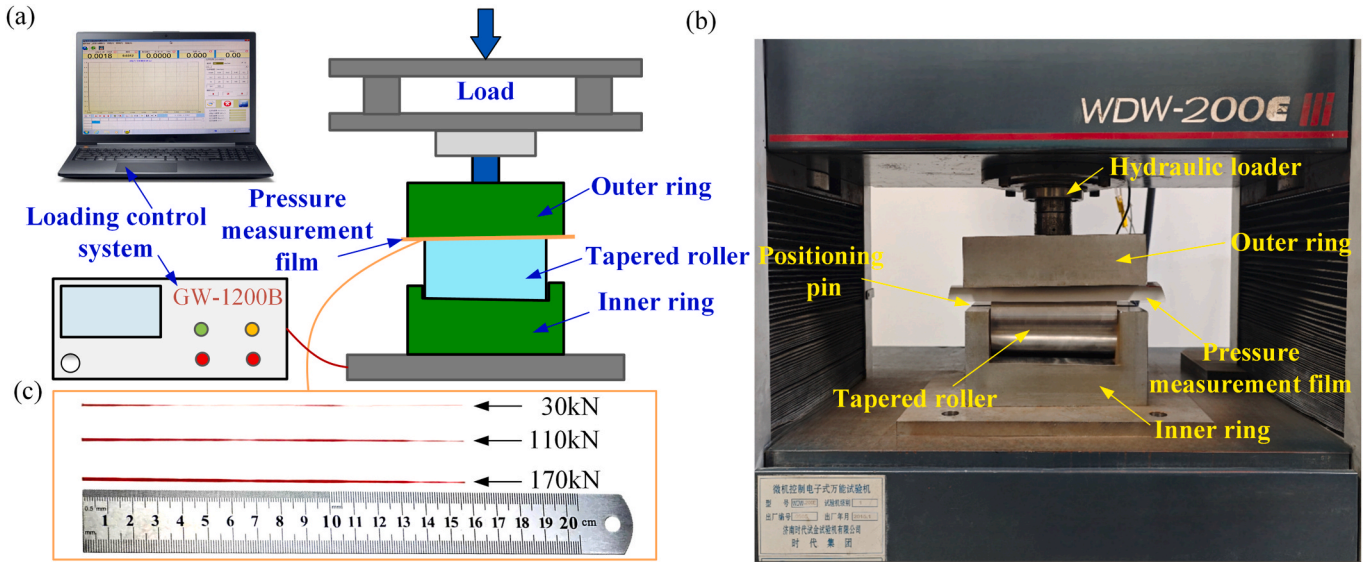


Fig. 11. (a) Schematic of the experimental setup, (b) Photograph of the experimental setup, and (c) Experimental indentation results.

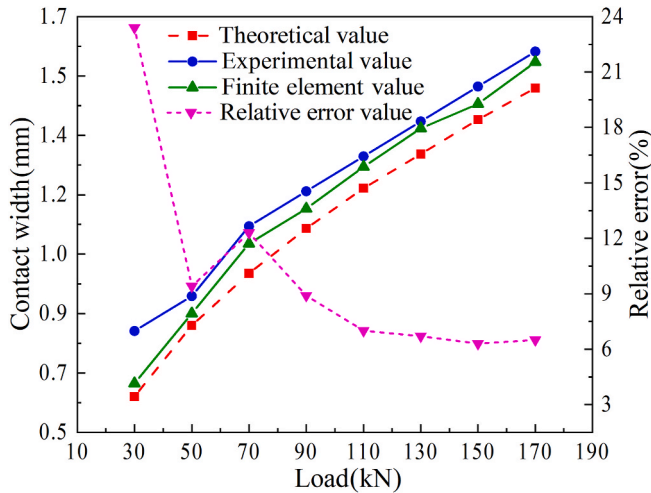


Fig. 12. Comparison of theoretical, finite element, and experimental values of contact width.

Table 2
Structural data of the main shaft transmission system.

L_1 (mm)	L_2 (mm)	L_3 (mm)	L_4 (mm)	G_1 (N)	G_2 (N)	γ (°)
2700	3000	5300	8000	1.7×10^5	4.5×10^5	8

Table 3
Typical load cases of the wind turbine.

	M_y (N.mm)	M_z (N.mm)	F_x (N)	F_y (N)	F_z (N)
Case A	0	1.2×10^9	0	1.7×10^6	0
Case B	4.8×10^9	1.2×10^9	0	1.7×10^6	0
Case C	4.8×10^9	1.2×10^9	0	1.7×10^6	1.9×10^6
Case D	4.8×10^9	1.2×10^9	3.6×10^6	1.7×10^6	1.9×10^6

radial component, offsetting a portion of the radial load. The main shaft DTRB structure enhances the ability of the system to resist overturning moments. Based on the mechanical model of the main shaft DTRB proposed in Section 2, the load-carrying state of the wind turbine main shaft DTRB under various load cases can be analysed.

4.2. Safety performance assessment

4.2.1. Structural strength analysis

Wind turbine manufacturers typically obtain extreme load data at the hub centre through simulations using the GH-Bladed aerodynamic software. The extreme load case with the maximum composite bending moment M_{yz} is selected (Table 4). According to the structural data of the main shaft transmission system, Eq. (13) is used to obtain the equivalent load of the main shaft DTRB. As shown in Fig. 14, to verify the accuracy of the finite element model, the contact load distribution of the fore bearing under extreme load conditions is calculated based on the Sjövä's integral method, the Harris model, ISO/TS 16281, and the finite element model. The contact load values calculated using the finite element model are lower, and the load-carrying region is larger. This is owing to the finite element model considering the elastic deformation of the bearing structural components. The maximum error between the theoretical and finite element contact load values is 6.3 %, indicating that the finite element method yields accurate results.

The contact stress distribution of the main shaft DTRB under extreme load conditions is then determined based on the finite element system model of the main shaft DTRB presented in Section 3.1, as shown in Fig. 15.

As shown in Fig. 15(a), stress concentration problems exist at both ends of the contact region of the fore bearing roller and they are more significant near the side of the rear bearing. As shown in Fig. 15(b), the contact stress between the roller and the raceway of the rear bearing is higher, and a significant stress concentration problem exists at both ends of the roller, which is more evident near the side of the front bearing. This is caused by the overturning of the main shaft owing to overturning moments M_y and M_z in the y and z directions at the wind turbine hub.

4.2.2. Effect of operating clearance on contact characteristics

The wind load is complex and fluctuating (Hart et al., 2019). To prevent the overload failure caused by the long-term stress concentration of the main shaft tapered roller bearing, accurate analysis of the contact between the roller and the raceway is necessary. Therefore, based on the mechanical model presented in Section 2 and the refined finite element sub-model presented in Section 3, a modification design is performed to consider the influence of operating clearance for the main shaft DTRB roller. The contact characteristics of the maximum load-carrying roller of the fore bearing under the extreme load case are shown in Fig. 16.

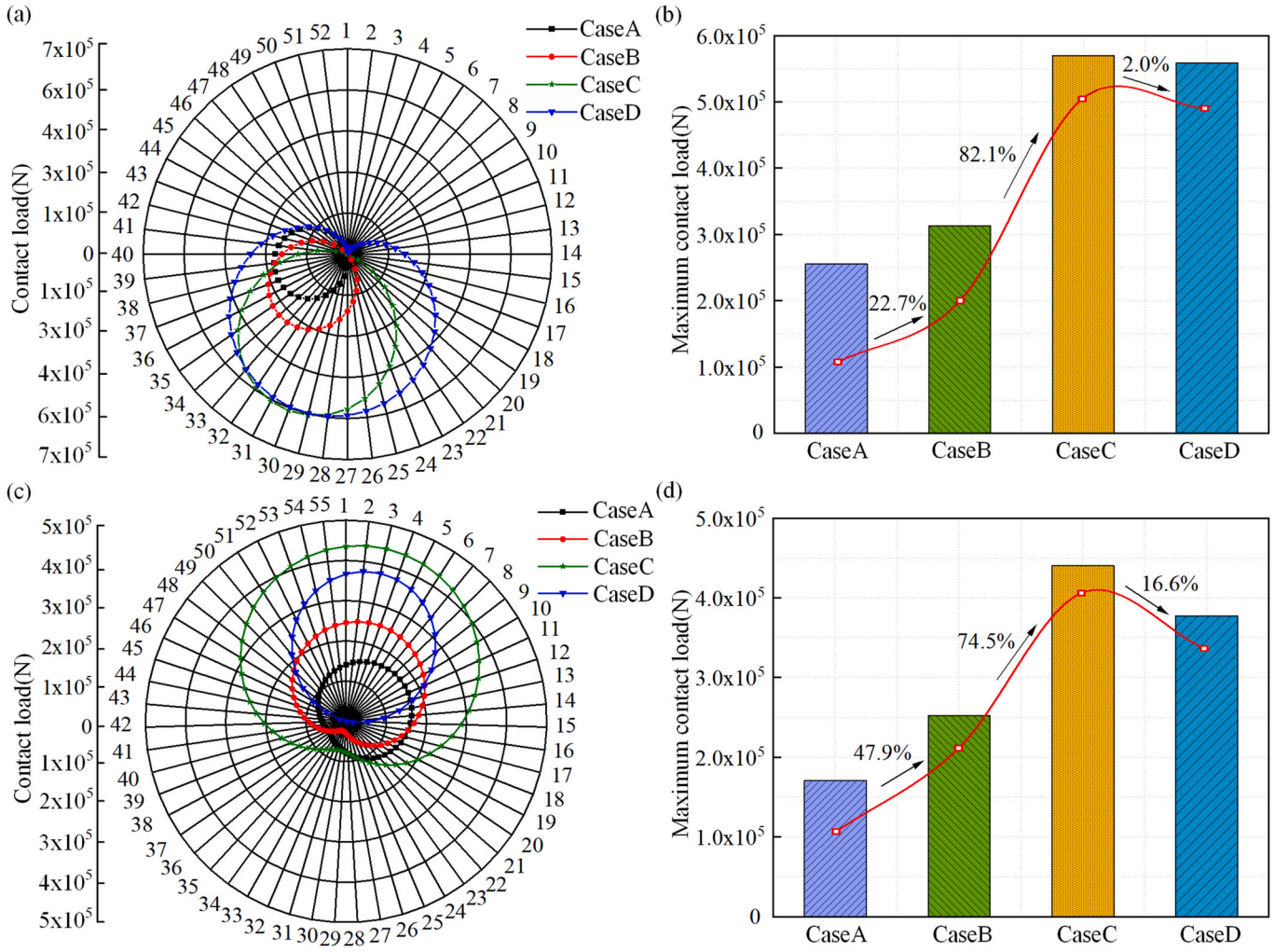


Fig. 13. Load distribution of the main shaft DTRB under typical load cases: (a) Fore bearing, (b) Maximum load value of the fore bearing, (c) Rear bearing, and (d) Maximum load value of the rear bearing.

Table 4

Extreme load case of the wind turbine.

M_y (N.mm)	M_z (N.mm)	F_x (N)	F_y (N)	F_z (N)
2.06×10^{10}	1.28×10^9	6.94×10^5	1.77×10^6	1.99×10^5

As shown in Fig. 16(a)–(b), when the radial clearance of the fore bearing is $150 \mu\text{m}$, the maximum contact stress between the unmodified roller and the raceway along the contact length is 3058.2 MPa. The maximum contact stress between the roller and the raceway gradually reduces as the radial clearance decreases. When the radial clearance is $0 \mu\text{m}$, the maximum contact stress between the roller and the raceway is 3017.9 MPa. When the radial clearance decreases to $-150 \mu\text{m}$, the contact stress reaches a minimum of 2971.2 MPa. However, as the bearing continues to be preloaded, the radial clearance is further reduced to $-250 \mu\text{m}$ and the maximum contact stress is increased to 3005.3 MPa. The radial clearance of TRBs directly impacts the load-carrying performance, and a significant stress concentration problem arises at both ends of the unmodified roller. The logarithmic modification of the roller effectively reduces the peak stress at both ends of the roller, as shown in Fig. 16(c). The maximum contact stress between the roller and the raceway is effectively reduced to 2547.9 MPa when the radial clearance is $0 \mu\text{m}$. The maximum contact stress at the radial clearance of $-150 \mu\text{m}$ reaches a minimum of 2515.1 MPa. TRBs typically control operating clearance by applying preload. The application

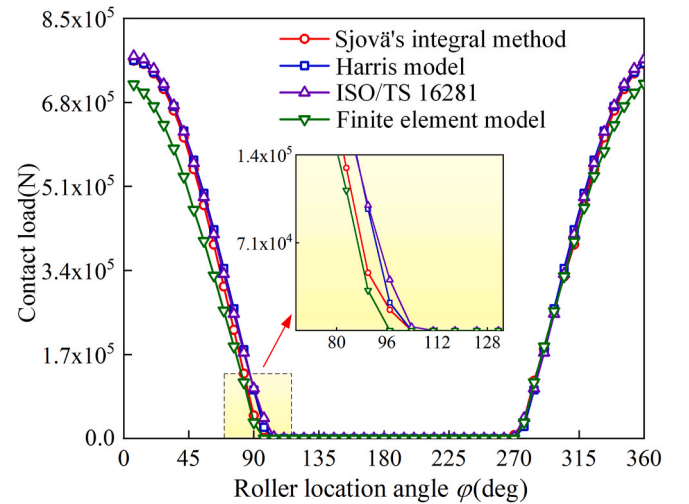


Fig. 14. Contact loads between the roller and outer raceway.

of appropriate preload during installation directly affects the performance and lifespan of the bearing. As shown in Fig. 16(d), when the preload is within range a , the maximum contact load between the roller and the raceway remains unchanged. When the preload increases to

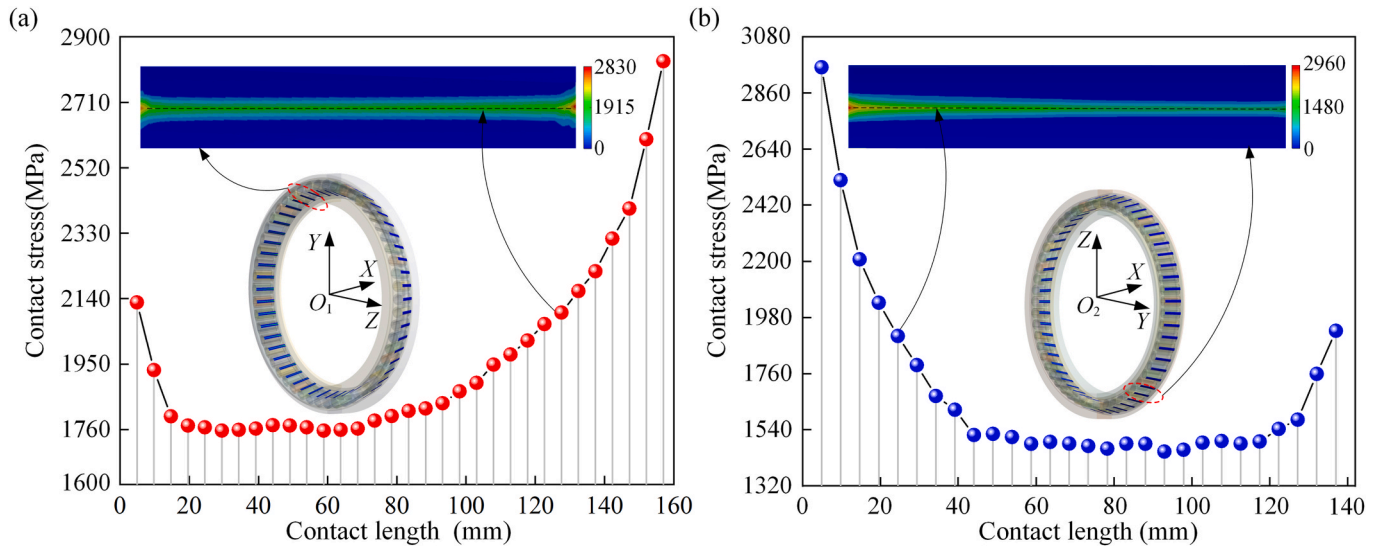


Fig. 15. The contact stress distribution of the main shaft DTRB: (a) Fore bearing, and (b) Rear bearing.

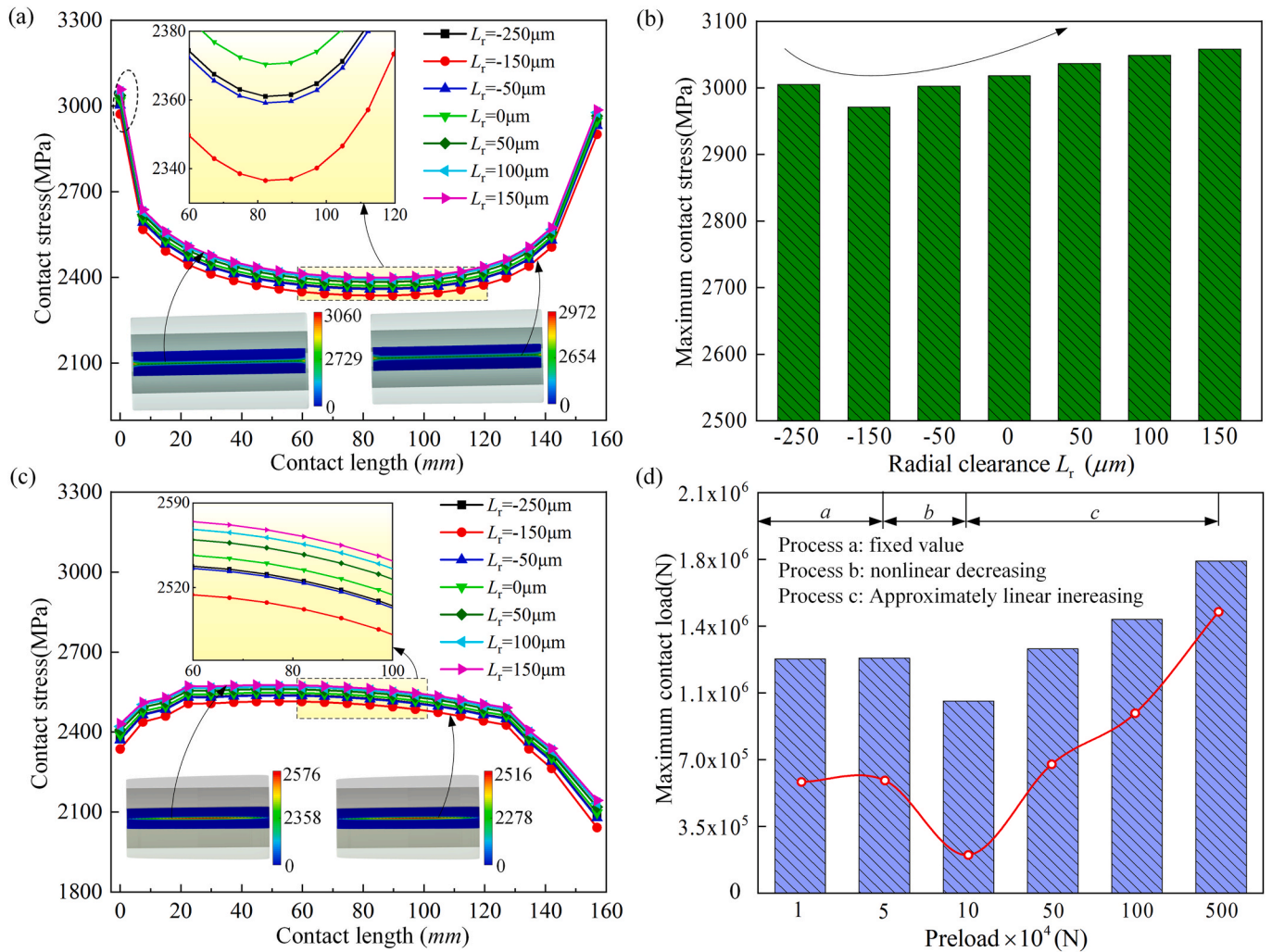


Fig. 16. Contact characteristics of the fore bearing under extreme load case: (a) Unmodified roller, (b) Maximum contact stress of the unmodified roller under different radial clearances, (c) Modified roller, and (d) Maximum contact load of the roller under different preloads.

range b , the maximum contact load reaches 1.01×10^6 N, showing a non-linear reduction of 18.3 % compared to range a . As the preload increases to range c , the maximum contact load increases almost linearly, detrimental to improving the load-carrying performance of the bearing.

The static load safety factor of the main shaft DTRB is calculated as $S_0 = [\sigma]/\sigma_{max}$, where $[\sigma]$ is the allowable Hertz contact stress and σ_{max} is the maximum contact stress between the roller and the raceway. By modifying the design of the main shaft DTRB roller, the static load safety factor of the rear bearing under the extreme load case is increased from 1.33 to 1.57: this is an improvement of 18.0 %. Similarly, the static load safety factor of the rear bearing is increased from 1.35 to 1.78, reflecting a significant improvement of 31.9 %. The roller modification effectively mitigates contact stress concentration between the rollers and raceways of the main shaft DTRB, thereby significantly enhancing safety performance.

4.3. Fatigue life assessment

4.3.1. Construction of the fatigue load spectrum

The fatigue life of the main shaft DTRB in a wind turbine is calculated using the load duration distribution (LDD) method (Germanischer, 2010). First, the time-series load data in the GL coordinate system at the wind turbine hub centre is processed using the rain-flow counting method (Zheng and Chen, 2022). The time-series load data includes force and moment components, such as F_x , F_y , F_z , M_y , and M_z . The process of constructing the fatigue load spectrum for the wind turbine MTRB is shown in Fig. 17.

As shown in Fig. 17(b), the LDD load spectrum includes 16 effective operating load conditions with different time proportions for the main shaft DTRB under rated conditions. Based on the LDD, the average equivalent bearing load level in both directions over the full life-cycle is calculated using Eq. (31). The load level is divided into 16 levels of fatigue load spectrum, as shown in Table 5.

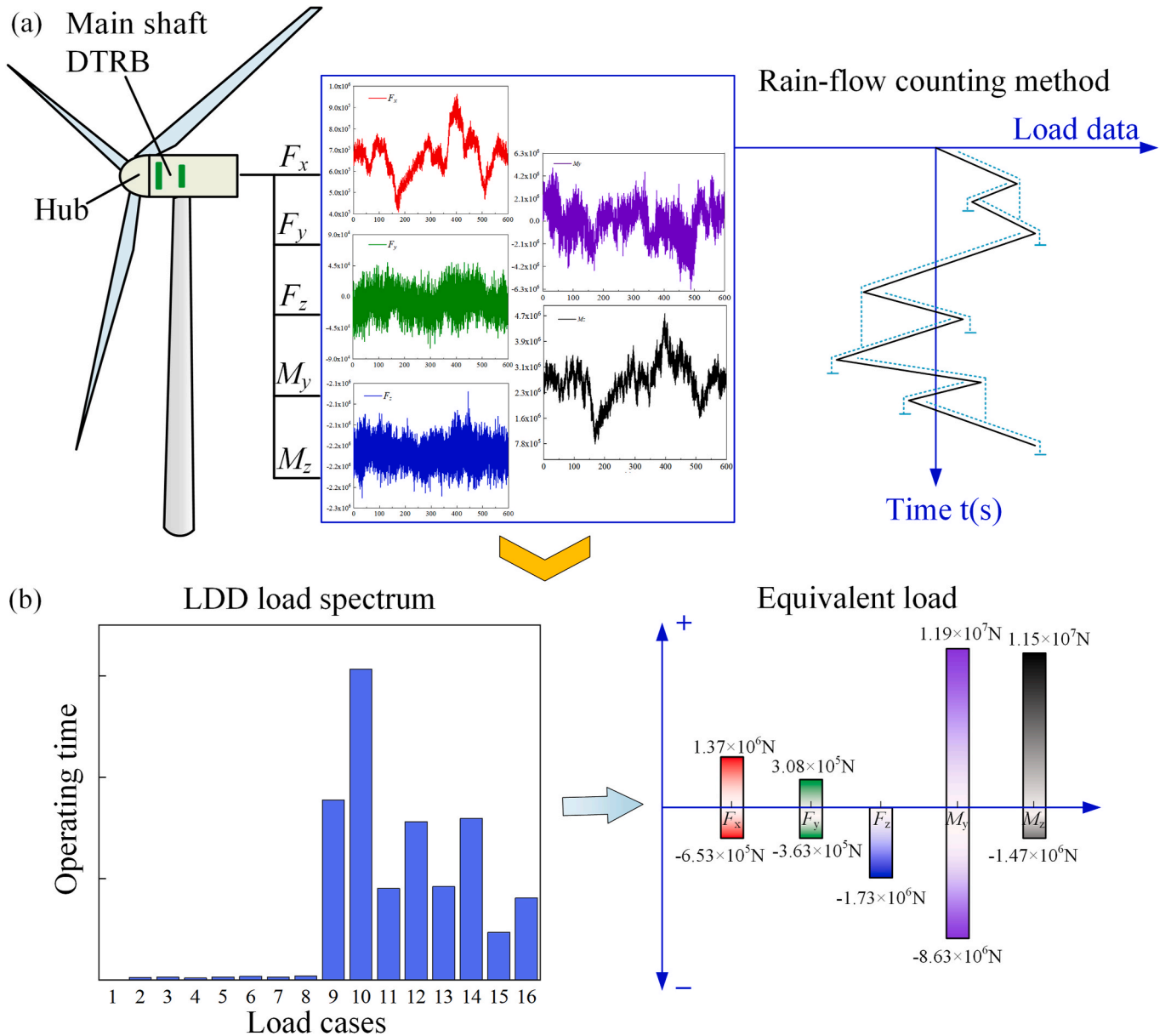


Fig. 17. Construction of the fatigue load spectrum for the main shaft DTRB: (a) Processing of time-series load data from the wind farm and (b) Calculating equivalent fatigue loads for the bearing.

Table 5
Fatigue load spectrum of the main shaft DTRB.

Load case	Time ratio q_i	F_x (N)	F_y (N)	F_z (N)	M_y (N·mm)	M_z (N·mm)
1	3.83×10^{-6}	-3.30×10^5	-2.93×10^4	-1.61×10^6	-1.68×10^6	-1.49×10^6
2	6.61×10^{-6}	-3.30×10^5	-2.93×10^4	-1.61×10^6	-1.68×10^6	1.74×10^6
3	1.95×10^{-6}	-3.30×10^5	-2.93×10^4	-1.61×10^6	2.27×10^6	-1.49×10^6
4	3.37×10^{-6}	-3.30×10^5	-2.93×10^4	-1.61×10^6	2.27×10^6	1.74×10^6
5	1.99×10^{-6}	-3.30×10^5	1.84×10^4	-1.61×10^6	-1.68×10^6	-1.49×10^6
6	3.44×10^{-6}	-3.30×10^5	1.84×10^4	-1.61×10^6	-1.68×10^6	1.74×10^6
7	1.01×10^{-6}	-3.30×10^5	1.84×10^4	-1.61×10^6	2.27×10^6	-1.49×10^6
8	1.75×10^{-6}	-3.30×10^5	1.84×10^4	-1.61×10^6	2.27×10^6	1.74×10^6
9	1.60×10^{-1}	8.65×10^5	-2.93×10^4	-1.61×10^6	-1.68×10^6	-1.49×10^6
10	2.76×10^{-1}	8.65×10^5	-2.93×10^4	-1.61×10^6	-1.68×10^6	1.74×10^6
11	8.14×10^{-2}	8.65×10^5	-2.93×10^4	-1.61×10^6	2.27×10^6	-1.49×10^6
12	1.41×10^{-1}	8.65×10^5	-2.93×10^4	-1.61×10^6	2.27×10^6	1.74×10^6
13	8.31×10^{-2}	8.65×10^5	1.84×10^4	-1.61×10^6	-1.68×10^6	-1.49×10^6
14	1.44×10^{-1}	8.65×10^5	1.84×10^4	-1.61×10^6	-1.68×10^6	1.74×10^6
15	4.23×10^{-2}	8.65×10^5	1.84×10^4	-1.61×10^6	2.27×10^6	-1.49×10^6
16	7.31×10^{-2}	8.65×10^5	1.84×10^4	-1.61×10^6	2.27×10^6	1.74×10^6

4.3.2. Fatigue life of the main shaft DTRB

Preloading is typically applied during the assembly stage to ensure the optimal operating condition of the wind turbine main shaft DTRB. The application of preload significantly affects the operating clearance of the main shaft DTRB. Therefore, this study focuses on analysing the influence of axial preload amount δ_{ax} on the fatigue life of the wind turbine main shaft DTRB system and determining a rational range for the preload amount, as shown in Fig. 18.

As shown in Fig. 18(a), with increasing preload amount δ_{ax} , the fatigue life of the fore bearing raceway initially increases and then decreases and the fatigue life of the rear bearing raceway decreases. An optimal δ_{ax} exists that maximises the fatigue life of the fore bearing. As

shown in Fig. 18(b), in the preload stage *a*, when the δ_{ax} reaches 150 μm , the fatigue life of the fore bearing raceway peaks at $1.96 \times 10^8 \text{ rev}$. In preload stage *b*, when the δ_{ax} reaches the range of 200–250 μm , the fatigue life values of the fore and rear bearings are nearly equal and both meet the permissible fatigue life requirements. As δ_{ax} increases, the fatigue life values of both the fore and rear bearing raceways decrease. In preload stage *c*, when the δ_{ax} exceeds 270 μm , the fatigue life of the rear bearing raceway falls below the permissible fatigue life values, failing to meet operational requirements. This preload range should be avoided to ensure optimal performance of the main shaft DTRB. Therefore, to ensure the proper operation of the main shaft DTRB and meet the permissible fatigue life requirements, using a preload δ_{ax} range of 150–270 μm . The operating clearance of the main shaft DTRB system of the wind turbine is set within the range of -270 to $-150 \mu\text{m}$ to achieve the optimal load-carrying state.

5. Conclusions

This study developed mechanical and finite element models for the main shaft DTRB of a semi-direct drive offshore wind turbine and assessed its fatigue life based on the fatigue load spectrum. The proposed mechanical model calculated the contact load distribution of the main shaft DTRB under various load cases. The refined finite element sub-model accurately solved the contact stress distribution between the rollers and the raceways. Experiments validated the effectiveness of the method. Finally, the influence of preload amount on the fatigue life of the main shaft DTRB was analysed based on the full life-cycle fatigue load spectrum. The conclusions are summarised as follows:

1. A mechanical model for the main shaft DTRB of a semi-direct drive offshore wind turbine was developed to analyse its load-carrying state under various load cases. The effects of bearing clearance and preload on the contact characteristics between the rollers and raceways were investigated. The results showed that appropriate preload and bearing clearance can significantly enhance load-carrying performance, providing a theoretical basis for understanding the failure mechanisms of the main shaft DTRB in offshore wind turbines.
2. The finite element model of the main shaft DTRB accurately analysed the contact characteristics between the rollers and raceways under external wind loads. The errors between the finite element results, theoretical values, and experimental data were less than 10 %, confirming the accuracy of the results. The structural strength of the main shaft DTRB under extreme conditions was analysed, showing that roller modification can effectively reduce stress concentration

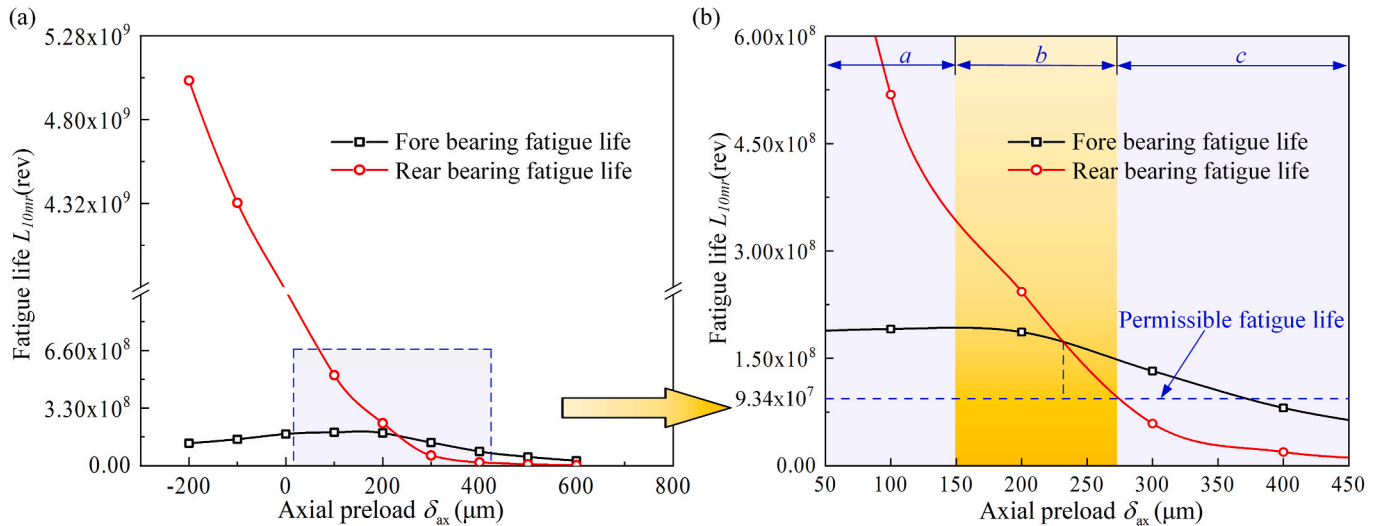


Fig. 18. Effect of axial preload amount δ_{ax} on the fatigue life of the main shaft DTRB.

between the rollers and raceways, thereby improving safety performance.

3. The full life-cycle fatigue load spectrum of the main shaft DTRB is derived based on the LDD and the Palmgren–Miner theory. The influence of preload amount on the fatigue life of the main shaft DTRB was analysed. The study found that the operating clearance range of the main shaft DTRB system is between -270 and -150 μm , ensuring both fore and rear bearings meet the 25-year fatigue life requirement. The study provides methods and guidance for the reliable operation and preload application of the DTRB system. However, the current fatigue analysis is only applicable to wind load environments with stable wind speeds, where there are no sudden high-stress events or complex loading sequences. Future work will further improve the analytical model to enhance the accuracy of fatigue life assessment.

CRedit authorship contribution statement

Lu Ren: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jianmei Wang:** Writing – review & editing, Supervision, Funding acquisition. **Yanwei Yuan:** Validation, Software. **Bo Wang:** Writing – review & editing, Supervision. **Dunmiao Wang:** Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper, and manuscript is approved by all authors for publication.

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Data availability

Data will be made available on request.

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